PROJECT OBJECTIVES

REVA Internship Opportunities (RIO) | School of Computer Science and Engineering, REVA University, Bengaluru

Particulars	Details
Task	Project Objectives, derived from Section 3 (Literature Gaps) of the Literature Survey Report
Name of Student	R Sai Pranav
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Project Title	ForgeGuard: An AI-Based Document Forgery Detection and Verification System
Track	PS 109
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1. Introduction

1.1 Purpose. This document converts the nine literature gaps identified in Section 3 of the Literature Survey and Requirement Analysis Report into a fixed set of project objectives. Each objective names the gap it closes and the requirement identifiers (Section 4 of that report) through which it is implemented, so objectives, requirements and the System Design Document all trace against the same identifiers.

1.2 Primary objective. To build ForgeGuard, a system that accepts a scanned or photographed document and returns an explainable, evidence-based authenticity decision — by combining signature verification, pixel-level tamper localisation, OCR/field validation and a fused authenticity score — rather than a single opaque genuine/forged label.

2. Objectives Derived from the Literature Gaps

Each objective below is numbered O1–O9 to match Gap 1–Gap 9 in the Literature Survey Report.

#	Objective	Closes / Implemented via
O1	Detect and clean the signature region directly on the uploaded page, instead of assuming a pre-cropped, clean signature image.	Gap 1 → FR-S1, FR-S2
O2	Verify a questioned signature against 1–5 stored references using a writer-independent similarity model, so a new signer can be verified with no retraining.	Gap 2 → FR-S3, NFR-6

O3	Train and evaluate tamper localisation on document images (not natural photographs), producing a pixel-level mask rather than a whole-image label.	Gap 3 → FR-D1, FR-D2
O4	Reduce reliance on JPEG/compression artefacts by fusing a noise-residual (SRM) stream with a semantic RGB stream, using ELA only as one supporting signal.	Gap 4 → FR-D1, FR-D2
O5	Fuse visual, signature and OCR/arithmetic evidence into a single authenticity score, so forgeries invisible to any one modality are still caught.	Gap 5 → FR-D3, FR-D4, FR-D5
O6	Make every decision explainable to a non-expert verifier: a 0–100 score, a label, a heatmap, per-check sub-scores and a plain-language reason.	Gap 6 → FR-D5, FR-D6, NFR-4
O7	Evaluate every model with precision, recall, F1 and AUC on held-out data, and handle class imbalance explicitly, rather than quoting accuracy alone on small datasets.	Gap 7 → NFR-5
O8	Keep each detector behind a fixed, replaceable interface, so an anomaly detector or a newer model for an unseen forgery type can be added later without redesigning the pipeline.	Gap 8 → NFR-7
O9	Maintain a tamper-evident, append-only audit record (document hash plus result) for every verification, so the system's own decisions stay checkable — and the design stays ready for blockchain anchoring in Phase 2.	Gap 9 → FR-P2, FR-A3, NFR-8

3. Scope Boundary

Objectives O1–O9 define Phase 1 and are the ones the System Design Document implements. Two items raised by the survey are explicitly out of scope for Phase 1: full blockchain anchoring of the document hash (papers [50], [51]) and hyperspectral ink analysis (paper [45]), both of which need hardware or infrastructure beyond a single web service. O9 keeps the design ready for the former by storing the hash now.

4. Conclusion

These nine objectives are exhaustive with respect to the gaps found in the survey: every gap in Section 3 maps to exactly one objective, and every objective maps to the requirement identifiers carried into the System Design Document. No objective is added without a corresponding gap, and no gap is left without an objective.